Ecological Inference and Racially Polarized Voting

A Georgia county where the inference can be checked against the ballots

84-355 Democracy's Data

August 2026

A federal judge decides whether a districting plan stands by reading a percentage in an expert report. In almost every county in America that percentage cannot be checked — not by the opposing expert, not by the judge, not afterwards by anyone. There is no held-out sample, no follow-up survey, no later disclosure. **The evidence on which American electoral maps are drawn is structurally unfalsifiable.**

In one kind of county it can be checked. This chapter is about what happens when it is.

01 Section 2 turns on a factual question

Section 2 of the Voting Rights Act turns on a factual question: **do voters of different races prefer different candidates?** If they do not, there is no claim. *Thornburg v. Gingles*, 478 U.S. 30 (1986), made it the second and third of the three preconditions a plaintiff must establish. That the minority group votes together, and that the majority votes as a bloc often enough to defeat the minority's preferred candidate.

> **The law under all of this changed on 29 April 2026.**
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> *Louisiana v. Callais*, 608 U.S. ___ (2026), did not overrule *Gingles*. It "update[d]" it, and the update lands squarely on the two preconditions this chapter is about. To satisfy them a plaintiff must now "provide an analysis that controls for party affiliation" and "show that voters engage in racial bloc voting that cannot be explained by partisan affiliation" (slip op., at 30). "[S]imply pointing to inter-party racial polarization proves nothing," the Court said, because a State may gerrymander for party "even if it so happens that the most loyal Democrats happen to be black Democrats." Five weeks later the Court restated it for a lower court that had missed it: "the mere fact that voters of different races vote for different parties is not relevant to proving racially polarized voting patterns," *Allen v. Milligan*, 608 U.S. ___ (2026) (per curiam) (No. 25A1314, June 2, 2026).
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> **The quantity estimated below is which party's ballot each racial group asked for.** That is the inter-party comparison, in its purest available form. Nothing about the estimation problem changes — the ballot is still secret, the precinct aggregates are still all anyone gets, and the methods still fail in the ways the rest of this chapter shows. What changed is what the answer is worth in court, and the closing sections say where that leaves it.

02 The ballot is secret

One obstacle stands in the way of answering it, and it was put there on purpose. **Nobody records how you voted.** What is recorded is how a *precinct* voted, and separately, who *lives* in that precinct. From those two aggregates an expert witness must infer how each racial group behaved, and a federal judge then decides a case on the answer.

That inference is called **ecological inference**, and it is the thing every statistics course warns you about: reasoning from groups to individuals. Done badly it is the ecological fallacy. Whether a districting plan is struck down turns on numbers produced this way — not on whether polarization exists in principle, but on the specific estimated percentages an expert puts in a report.

03 Why this county is different

In Georgia two facts are public that are private nearly everywhere else. The race a voter reports on the registration form, and **which party's primary ballot they requested**. So for one election, the General Primary of 21 May 2024, the exact quantity ecological inference is trying to estimate is on record, person by person.

So this is one of the very few places where a guess about *what people did* can be graded against what they actually did.

04 The field has been arguing since 1953

The argument started when Goodman published the regression approach, which fits a best line through a scatter of points. Duncan and Davis published the bounds approach in the same volume of the same journal. King's 1997 book proposed a method combining them. Freedman and colleagues (1991) had already argued, in a paper written for voting-rights litigation, that the regression approach is not reliable enough for the use courts make of it. **None of these positions has won.** All of them are still being cited in briefs.

05 Three bureaucracies, none of which meant to build an answer key

The file underneath this chapter is a join of three separate public records, each collected by the State of Georgia for its own administrative reason and none of them for research. Race comes off the voter registration form. It sits there because the Voting Rights Act obliged covered states to be able to report who was registering. Party of the primary ballot comes from the voter-history file. It sits there because a state running partisan primaries has to record which one each voter took.

Precinct comes from a registration extract, where it sits so a voter can be sent to the right polling place. Put the three together for the General Primary of 21 May 2024, and the quantity ecological inference exists to estimate is simply written down, voter by voter. **Nobody built that answer key. It is a by-product of three unrelated obligations.**

The copy used here is the working data assembled for the county's 2026 redistricting matter. It is the same underlying extract the surname-inference chapter grades BISG against.

That last sentence carries more weight than it looks like it does. These are public records, and "public record" in this context does not mean a link. A Georgia voter file is obtained by asking the state for an extract. It is cut on a particular day, given a number, and then never regenerated. The register itself keeps moving underneath it as people register, move and are removed.

So this file is a photograph of an electorate rather than a live feed. It came here through the work of an expert in a redistricting case rather than off a website. And **a reader cannot fetch it again to check the numbers below.** They would have to request their own extract, which would be a different one.

That is the ordinary condition of voter-file research. It is worth naming, because nothing about the tidy 17-row file kept here gives it away.

It is the right evidence for grading a method and the wrong evidence for settling anything about this county's politics, and the distinction is worth being pedantic about. The recorded quantity is *which party's ballot a voter asked for in a May primary*. That is not the same as which candidate they supported in November, and a primary electorate is not the general electorate.

What would have been better is a general election with the same answer key. That file cannot exist anywhere, because the thing that makes November's ballot secret is the whole point of the secret ballot. So the chapter grades the method on the one election where grading is possible, and says so rather than quietly generalising.

There is a second reason this quantity is worth having, and it belongs to the box above. The party of a requested ballot is the between-party comparison in its purest available form. That is exactly the comparison *Callais* now says proves nothing on its own.

Three exclusions were applied before the file was written and all three move the numbers. Voters whose race had been **filled in by BISG** rather than reported by the voter were dropped. Grading one guess against another would have produced a beautiful and meaningless result. It is the same trap the surname chapter disarms.

A few hundred voters took a ballot without requesting either party's. They are counted as neither Democratic nor Republican, rather than being assigned to one.

And the analysis is restricted to voters who reported Black or white. That is standard in Section 2 practice, and it is what makes a two-column method work at all. The people it removes voted, and the chapter returns to what their absence costs.

The last piece of cleaning is the one that shapes what you can see. **No individual record is committed here.** The registration file carries fifty-odd columns, including name, birth year and street address. The vote-history file has one row per person. All of it was collapsed to 17 precinct rows before anything was saved.

That is a deliberate privacy decision with a cost, and the cost is exactly the subject of this chapter. From here on, this page has what an expert witness would have, plus two extra columns holding the answer.

06 The join, shown once, because after this it is gone

Three files, three offices, no shared purpose, and one thing in common: a registration number. Here is what each contributes, and what survives each step.

Step	Rows
Vote-history rows, 21 May 2024, D or R ballot	17805
Self-reported race, imputation removed, Black or white	132452
Registration extract rows carrying a precinct	127560
Joined on registration number, all three present	16024
Precinct rows written	17

Read those counts downward and the join is doing something drastic. The vote-history file offers 17,805 people who took a party ballot on 21 May. The race file offers 132,452. The registration extract offers 127,560.

What comes out the other side is 16,024, because a voter has to be in **all three**. They must have voted in that primary. They must have reported a race rather than had one imputed, meaning guessed at from other things. And they must still be in the current registration extract with a precinct attached.

Each of those three conditions is a judgment, and each removes real voters. The primary condition is the chapter's subject and cannot be relaxed. The imputation condition is deliberate. Grading one guess against another would produce a beautiful and meaningless result.

The extract condition is the one nobody chooses. It removes anybody who has moved or died since May 2024. That is the same silent decay the registration chapter measures, arriving here as a smaller denominator, a smaller bottom number to divide by.

The joined record is the thing ecological inference exists to reconstruct, so it is worth seeing once. These are four real voters, with the registration number replaced by a row label; no name, address or number is here.

Voter	Party	Race self	Precinct
voter 1	REPUBLICAN	White	ROZR
voter 2	REPUBLICAN	White	CENT
voter 3	DEMOCRAT	White	HHPC
voter 4	REPUBLICAN	White	MCMS

That is all there is to it. **Race, party, precinct — three columns, one voter per row, and the question this whole chapter is about is already answered.** Count the rows where race is Black and party is Democratic. Divide by the rows where race is Black. That is the number eight pages of method are about to estimate badly.

Now watch it disappear:

Precinct	Voters	Black	White	Dem	Rep
ROZR	1606	225	1381	262	1344
WELLSTON CENTER	772	591	181	595	177

16,024 rows became 17. Every voter-level row was replaced by a set of **margins**: how many Black voters, how many Democratic ballots. The cross-tabulation that linked them is gone. Nothing in the top table can be recovered from the bottom one. That is not an accident of this build. It is the exact condition of every Section 2 case ever litigated, because the ballot is secret and the cross-tabulation is available to nobody.

Two columns here break that rule on purpose, and they are the reason the chapter can grade anything. `black_dem` and `white_dem` are the counts the margins cannot give you, carried forward from the join and set aside. They are withheld from every method in the next several sections and produced at the end as an answer key. **An expert witness in a real case has the table above and does not have those two columns**, which is the whole difficulty, stated as a data structure.

07 One row is a precinct

Precinct	Primary voters	Black	White	Democratic ballots	Republican ballots
WELLSTON CENTER	772	591	181	595	177

One row per precinct. Two columns matter for the inference: **what share of the precinct's primary electorate was Black**, and **what share of its ballots were Democratic.**

Quantity	Value
Precincts	17
Primary voters	16,024
Black voters	4,631
white voters	11,393
Democratic ballots requested	5,106
Republican ballots requested	10,918
Least Black precinct	HEFS (10.9% Black)
Most Black precinct	WELLSTON CENTER (76.6% Black)

This table is the entire evidentiary record. An expert witness in a Section 2 case gets essentially this and nothing more.

Note the last two rows. **No precinct in this county is overwhelmingly Black.** That matters more than it looks.

08 The eyeball

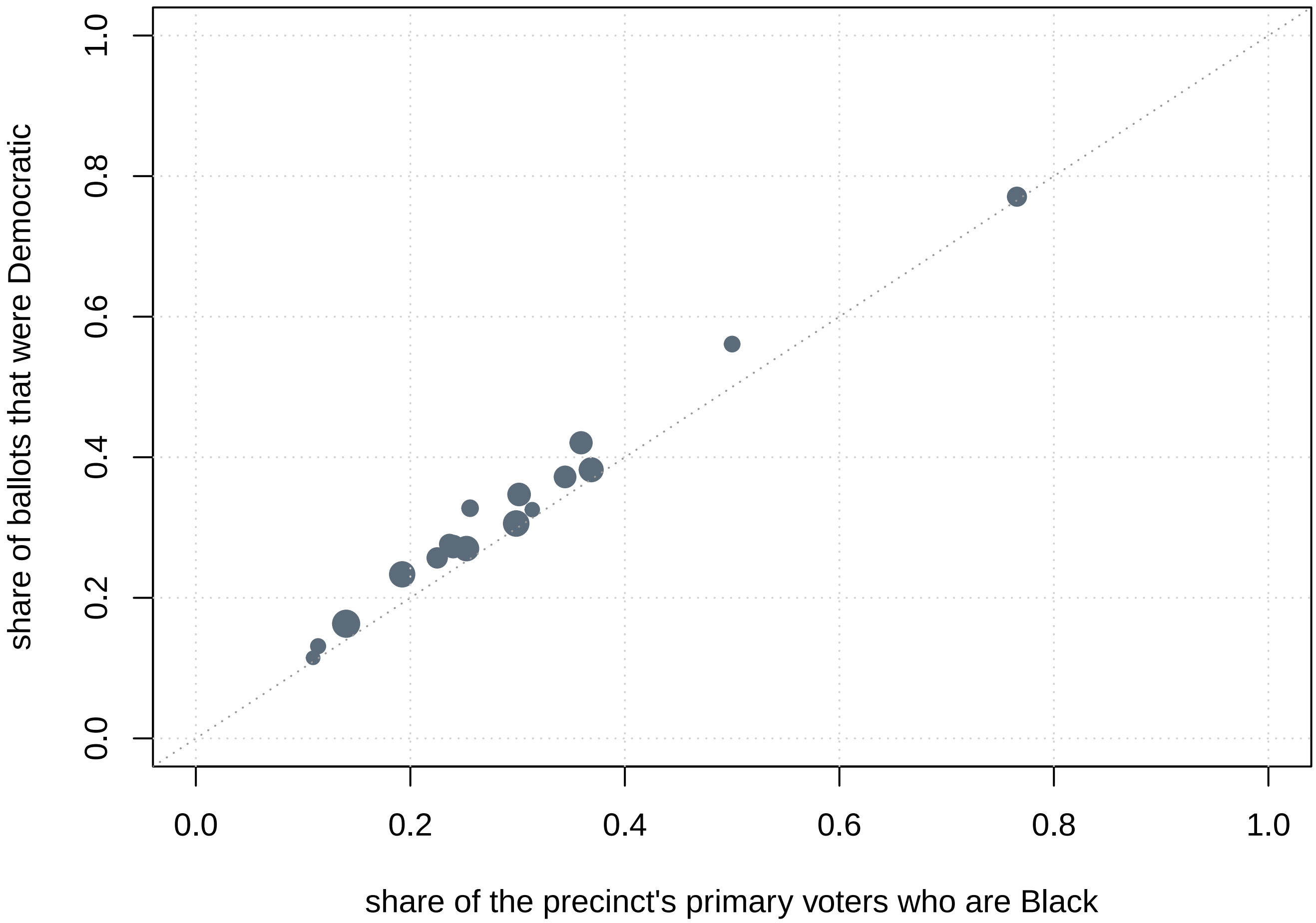


Figure 1. Seventeen precincts, race against ballot. One dot per precinct, sized by the number of primary voters. Each is placed by the share of its primary electorate that is Black, and the share of its ballots that were Democratic. The dashed 45-degree line is where the points would sit if every Black voter took a Democratic ballot and no white voter did.

The relationship is about as clean as social science gets, and the points sit close to the 45-degree line. That is what you would see if **every** Black voter took a Democratic ballot and **no** white voter did.

That is a hypothesis, not a finding. Two different worlds produce this picture: one in which Black and white voters are perfectly polarized, and one in which Democratic-leaning people of every race happen to live near each other. **The scatter cannot tell them apart, and neither can any method below.**

Since *Callais* there is a third world, and it is the one that now decides cases. Black and white voters here differ by **party**. The same picture would appear in any state where race and party go together, which is most of the South. Ruling that world out is the plaintiff's job now, and nothing plotted on this axis can do it. The y-axis *is* party.

09 The method courts trust most

The oldest approach in voting-rights litigation, and still the one judges find most intuitive: find precincts that are nearly all one race and look at how they voted. No modeling, no assumptions.

Quantity	Value
Precincts at least 80% Black	0
Precincts at most 20% Black	4
Democratic share in those white-homogeneous precincts	18.0%

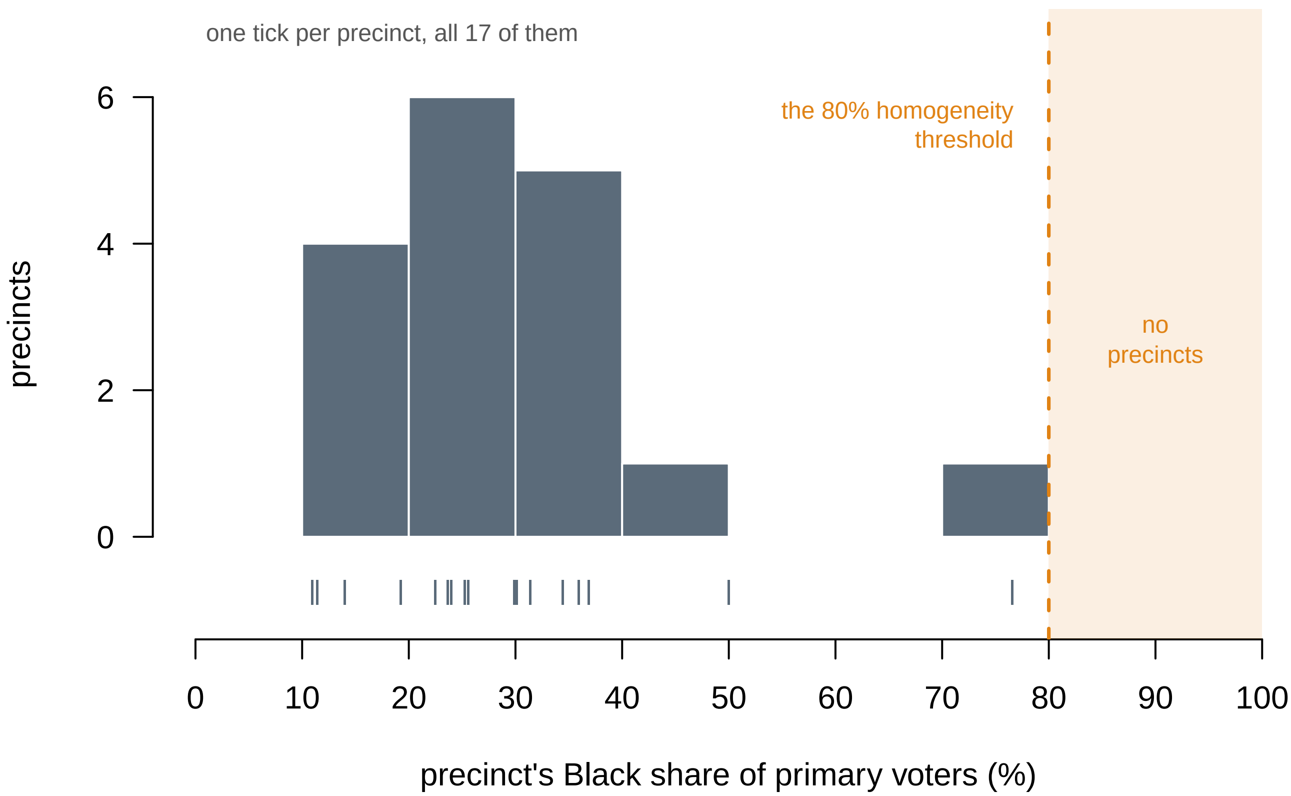


Figure 2. Where the county's precincts actually sit on the axis. Each bar is a ten-point slice of the precinct's Black share of primary voters; the ticks below the axis are the 17 individual precincts. The shaded strip is the 80% homogeneity threshold. The most Black precinct in the county is 76.6%, and 4 of the 5 decades above 50% contain nothing at all.

There are none. Not one precinct in this county reaches 80% Black. So the most trusted method in the field cannot produce an estimate for Black voters here at all. Not an imprecise one. None.

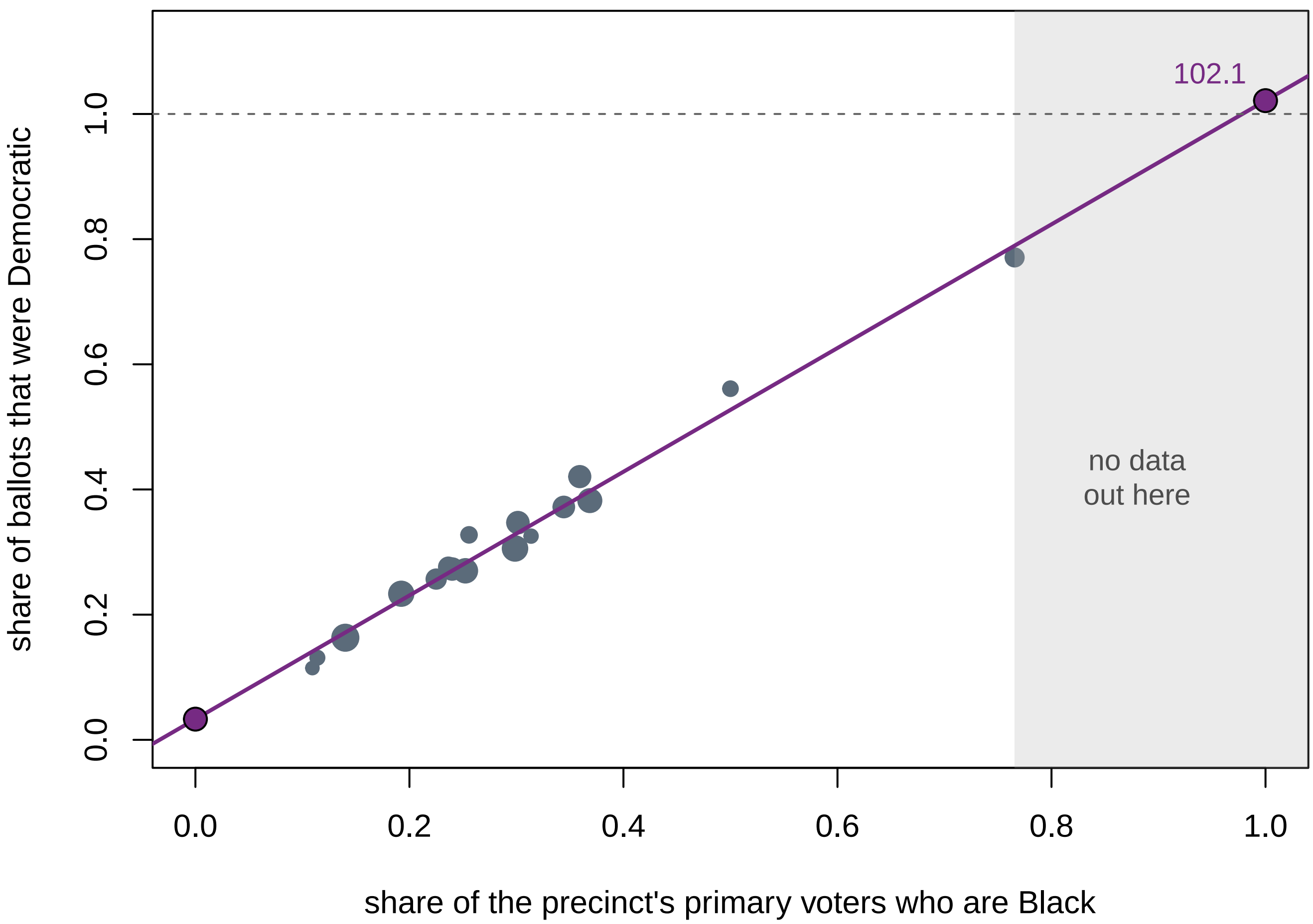
That is not an artifact of the threshold. It is a fact about the county, and it is the ordinary situation in most of the United States. **The method works best exactly where segregation is most severe.**

And the number it *does* produce deserves distrust. Those 4 "white-homogeneous" precincts are up to 20% Black, and those Black voters cast ballots too. The estimate for white voters is contaminated by the group being held out.

10 Goodman's ecological regression

The standard alternative, and for fifty years the workhorse of Section 2 litigation. Fit a line through the scatter and read off its ends. Where the line crosses zero percent Black is the estimate for white voters. Where it crosses one hundred percent is the estimate for Black voters.

Quantity	Value
R-squared	0.982
Estimated Democratic support, white voters	3.3%
Estimated Democratic support, Black voters	102.1%



The two large purple points are the estimates. The right−hand one sits in the gray band, where the county contains no precincts at all.

Figure 3. The fitted line, and where it has to go to produce an answer. The same precincts as Figure 1, with Goodman's least-squares line through them. The two large points are the estimates: where the line crosses 0% Black is the estimate for white voters, where it crosses 100% is the estimate for Black voters. The gray band is the part of the axis the county contains no precincts in — everything above 76.6% Black. The dashed rule at the top is 100%, the ceiling of possibility.

The model fits beautifully, with an R^2 of 0.982, and returns an impossible answer. It estimates that 102.1% of Black voters took a Democratic ballot.

There is no gentle reading of that. A percentage above 100 is not a close call or a rounding problem; it is a signal that the method has been asked to do something it cannot do. And **nothing in the regression complained.** The fit statistics are superb. An R^2 of 0.98 is the kind of number that gets quoted in a report.

The gray band shows why. The most Black precinct in the county is 76.6% Black, so the estimate for Black voters comes from extending the line 23 points past the last observation. The model is not summarizing data out there. There is no data out there.

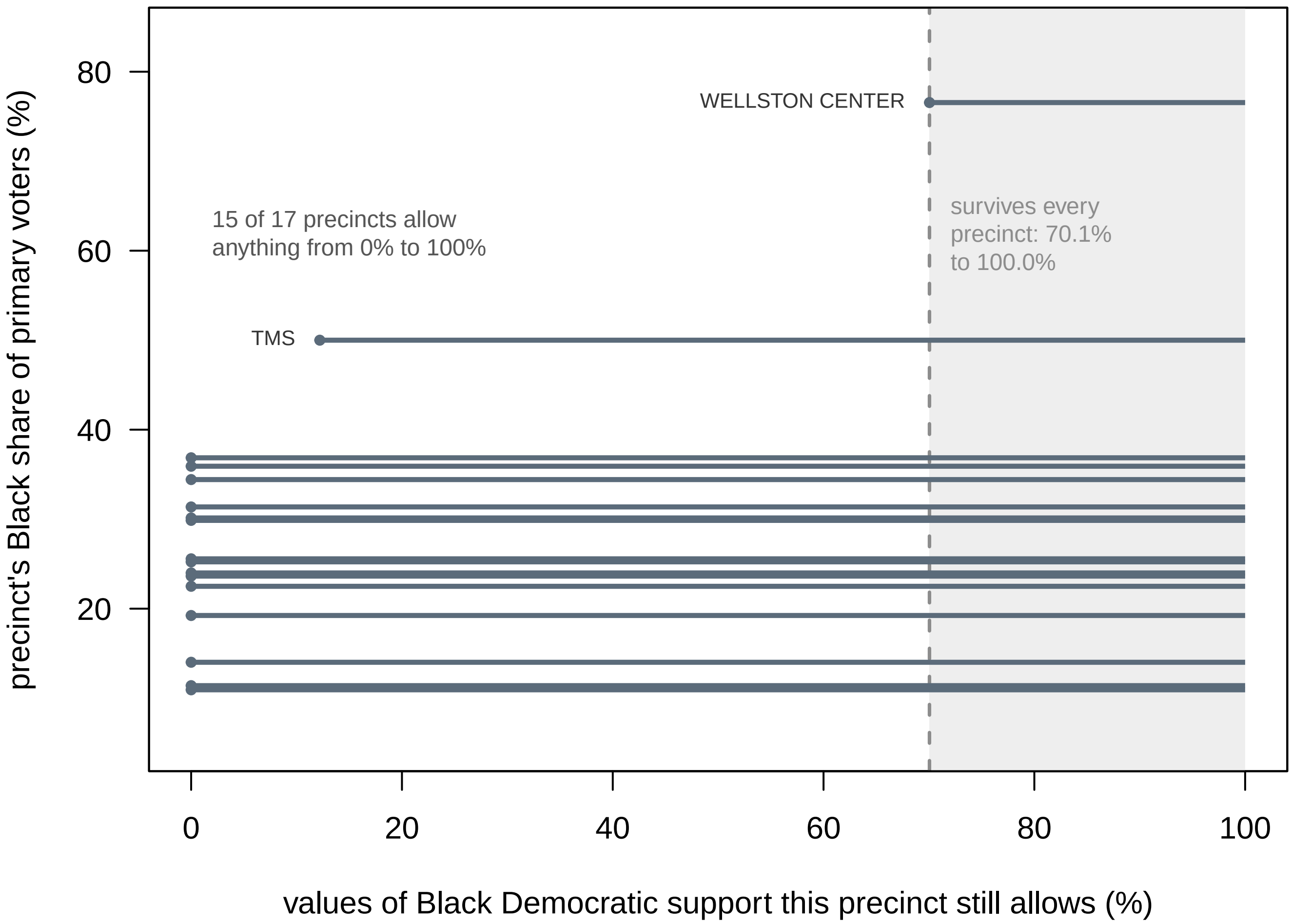
11 What is knowable without any model

Before King, Duncan and Davis pointed out something simple. Even with no model at all, the precinct totals *limit* the answer. Suppose a precinct is 40% Black and cast 45% Democratic ballots. Even if every white voter went Democratic, Black voters must supply the rest. That puts a floor and a ceiling on their behaviour.

Quantity	Value
Lower bound on Black Democratic support	70.1%
Upper bound	100.0%
Width of the interval	29.9 points
Which precinct produces the binding lower bound	WELLSTON CENTER

Black Democratic support is between 70.1% and 100.0%. That statement requires no assumptions whatsoever. It is arithmetic on the marginals and it cannot be wrong.

Every precinct produces its own interval, and the answer has to satisfy all of them at once. Here they are, stacked by how Black the precinct is.



One bar per precinct: the values of Black Democratic support that precinct's own totals still permit. 15 of the 17 precincts rule nothing out at all, and the shaded strip is what is left after every one of them.

Only 2 of the 17 precincts rule out anything at all. **Figure 4. What each precinct's own totals still permit.** One bar per precinct, stacked by how Black the precinct is: the bar spans the values of Black Democratic support that precinct's marginals leave possible. 15 of the 17 precincts rule nothing out at all. The shaded strip is what survives after every precinct's constraint is imposed at once — 70.1% to 100.0%.

The binding constraint comes from WELLSTON CENTER, the most Black precinct in the county — **the same polling place that a later section removes.**

It is also 29.9 points wide, which is why nobody testifies with it alone.

That is the trade the whole field runs on. The bounds are guaranteed correct and too vague to decide a case. The regression is precise, decisive and impossible. King's method is an attempt to get something in between: use the bounds as a hard constraint, then use a statistical model to say where inside them the answer probably sits.

> **Before you read on, write two things down.** There are now three estimates for Black Democratic support on the table: none at all from the homogeneous precincts, 102.1% from the regression, and "somewhere between 70.1% and 100.0%" from the bounds.
>
> First, **your own point estimate**. Second, and this is the question that matters: if a re-

port handed you an R^2 of 0.982, a tight scatter and a clean line, **how far off would you expect the estimate to be?**

12 The answer

Every voter in that table requested a party ballot by name and reported their race on the registration form.

Method	Black voters	White voters
Homogeneous precincts	cannot be computed	18.0%
Goodman's regression	102.1%	3.3%
Bounds, lower	70.1%	
Bounds, upper	100.0%	
THE TRUTH	92.7%	7.1%

Those two percentages are a summary of 16,024 individual decisions. Here they are as decisions: every Black and white voter in the county, flowing from the race on their registration form to the ballot they asked for.

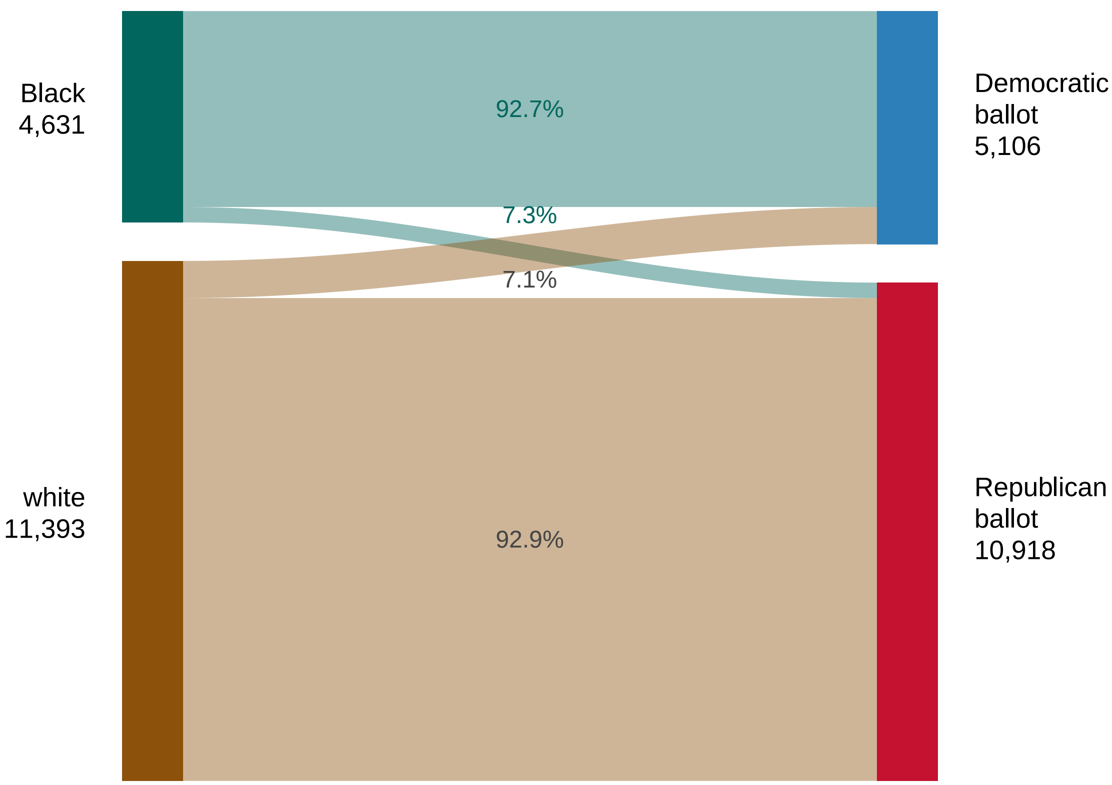
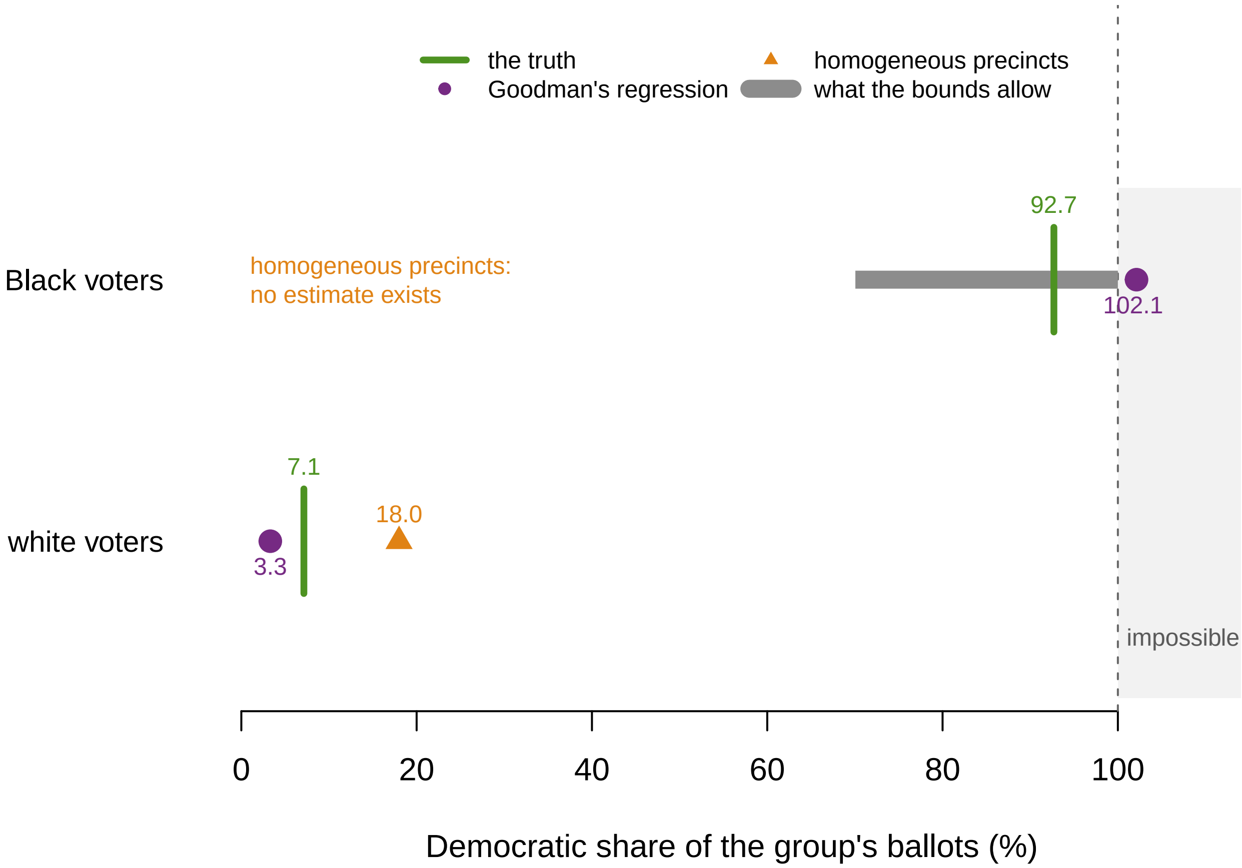


Figure 5. 16,024 individual decisions, from race to ballot. Ribbon thickness is the number of voters flowing from the race recorded on their registration form to the party ballot they asked for. Voters of every other race requested ballots in this primary too and ap-

pear nowhere in this diagram; they were dropped before the file was built, because the two-group framework has nowhere to put them.



Every estimate for both groups on one axis, against the recorded truth. The regression's Black estimate sits in the gray zone past 100%.

Figure 6. Every estimate, against the recorded truth. One row per group. The thick bar is the deterministic bounds; the dot is Goodman's estimate; the vertical rule is what the voter file actually recorded. The shaded zone past 100% is impossible, and the regression's estimate for Black voters sits inside it.

The truth is 92.7% for Black voters and 7.1% for white voters.

This electorate is about as racially polarized as American politics gets, and every method said so.

On the question these methods were built to answer, they all got it right. Is voting polarized here? Yes.

On the question *Callais* now asks, none of them took a position at all. Is it polarized by **race** rather than by **party**? They could not. The outcome recorded in this file is the party of the ballot requested.

Now look at how.

- **The bounds contained the truth,** as they must. They always do. That is their entire virtue and it cost 29.9 points of precision.
- **Goodman missed by 9.4 points** and landed outside the range of possible numbers. Its white estimate, 3.3%, is less than half the true 7.1%.

- **Homogeneous precincts could not answer at all** for Black voters, and overstated white Democratic support by 10.9 points.

The uncomfortable part is that **the method that was most confident was the one that was wrong, and its confidence was legible.** R^2 of 0.982, a tight line, a clean plot. Nothing in the output warned anybody.

The impossibility was the only warning, and only because the truth happened to sit near a boundary. In a county where Black Democratic support were 60%, Goodman could have missed by the same nine points and returned a perfectly plausible 51%, and nobody would ever have known.

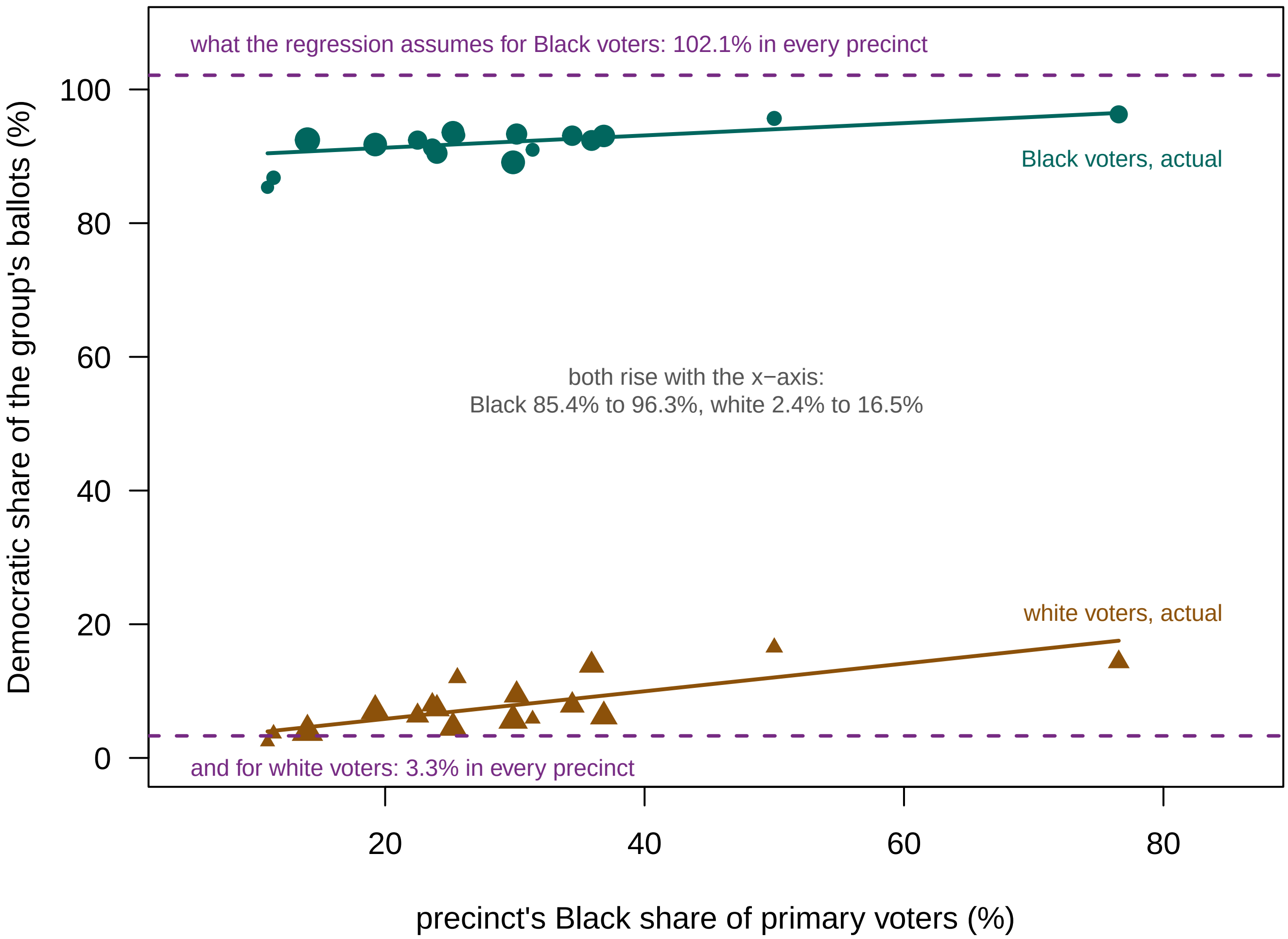
13 Why it failed, precisely

The failure is not that the model fits badly. Compare what it predicted for each precinct with what each precinct actually did.

Precinct	% Black	% Dem, actual	% Dem, predicted	Black Dem rate (true)	White Dem rate (true)
HEFS	10.9	11.5	14.1	85.4	2.4
HAFS	11.4	13.1	14.6	86.8	3.6
ROZR	14.0	16.3	17.1	92.4	3.9
BMS	19.2	23.3	22.3	91.8	7.0
CGTC	22.5	25.7	25.5	92.4	6.3
ANNX	23.6	27.6	26.7	91.3	7.9
PEC	24.0	27.3	27.0	90.5	7.3
VHS	25.2	27.0	28.2	93.6	4.5
VFW	25.6	32.7	28.6	93.2	12.0
MCMS	29.9	30.6	32.8	89.1	5.7
TWPK	30.1	34.7	33.1	93.3	9.4
NORTH HOUSTON SPORT COMPLEX	31.4	32.5	34.3	91.0	5.8
CENT	34.4	37.2	37.3	93.1	7.9
HHPC	35.9	42.1	38.8	92.4	13.9
FMMS	36.9	38.2	39.7	93.0	6.2
TMS	50.0	56.1	52.7	95.7	16.5
WELLSTON CENTER	76.6	77.1	79.0	96.3	14.4

The model predicts every precinct's Democratic share to within 4.2 points, and still misstates the group rate by 9.4. Predicting the total and decomposing the total are different problems. **That is the ecological fallacy, shown rather than asserted.**

The mechanism is in the fifth column. Goodman's regression assumes the Black Democratic rate is the *same in every precinct* — that is exactly what a single intercept and slope encode. In this county it runs from 85.4% to 96.3%, a spread of 10.9 points, and it is correlated 0.72 with the precinct's Black share.



Circles: how Black voters in each precinct actually voted. Triangles: how white voters in the same precinct actually voted. The regression assumes each is one number (the dashed lines); both slope upward with the x−axis instead.

Figure 7. The assumption the regression makes, and what was actually true. Circles are how Black voters in each precinct actually voted. Triangles are how white voters in the same precinct voted. Both are plotted against how Black the precinct is. The dashed rules are what Goodman's regression assumes: one number for each group, the same in every precinct.

Both true rates slope upward instead. The Black rate runs from 85.4% to 96.3%, and the white rate from 2.4% to 16.5%.

Black voters are more Democratic in precincts with more Black voters. That link between how a group behaves and how large it is locally is exactly the condition under which ecological regression breaks. It cannot be detected from totals. From outside, it just looks like a slightly steeper line.

14 How much rests on one polling place

WELLSTON CENTER is the only precinct above 50% Black, so it anchors the entire right-hand end of the line. Refit without it.

Fit	Black estimate	White estimate	R squared	Most black precinct
All 17 precincts	102.1%	3.3%	0.982	76.6%
Without WELLSTON CENTER	106.8%	1.7%	0.966	50.0%

Removing one precinct out of 17 moves the Black estimate from 102.1% to 106.8%, further into impossibility. The R^2 stays at 0.966.

Do it 17 times, once for each precinct, and the dependence is not subtle.

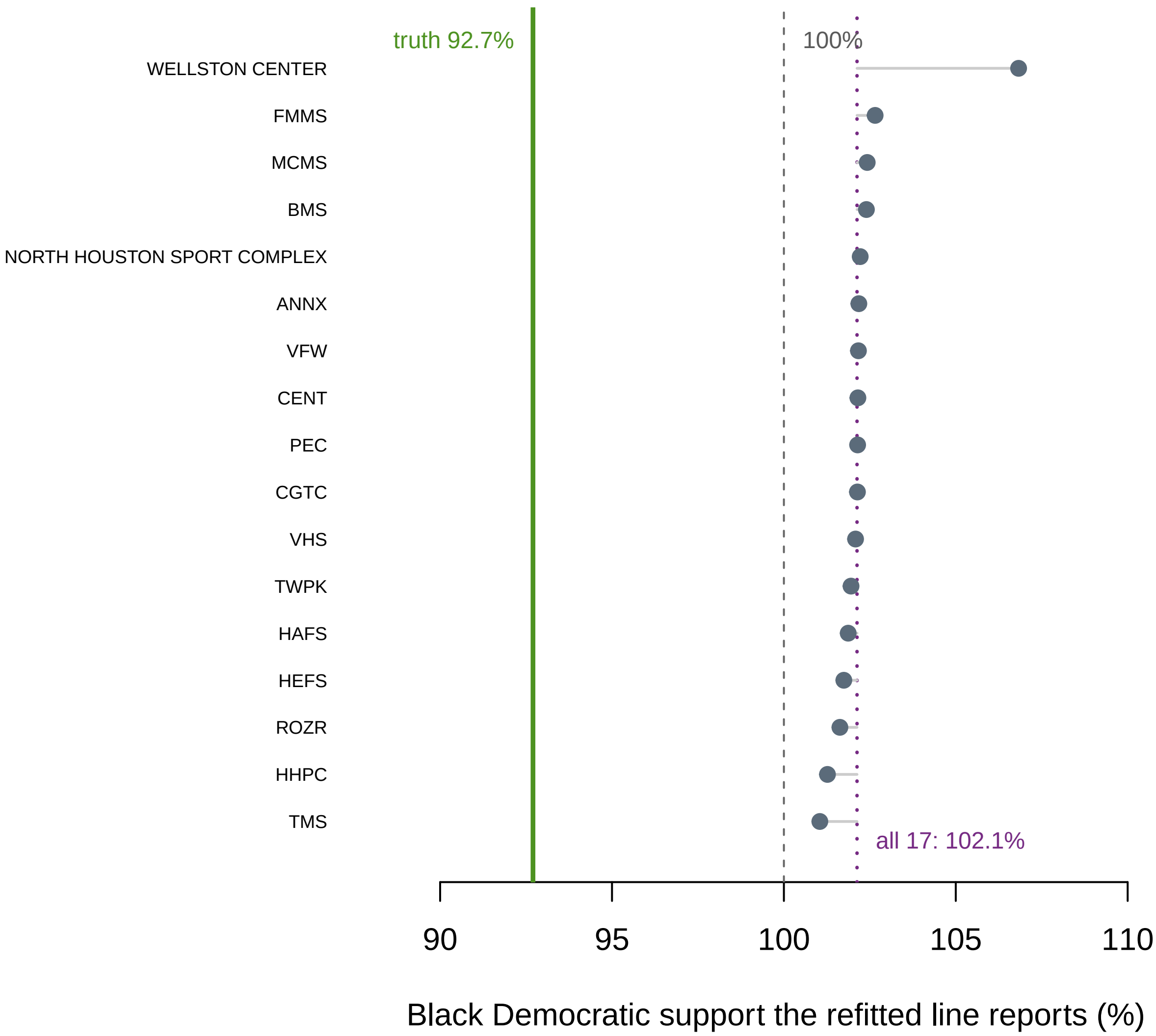


Figure 8. The line refitted 17 times, dropping one precinct each time. Each row is the Black Democratic estimate the regression returns when that precinct is left out. The estimate ranges over 5.8 points and stays above 100% in 17 of the 17 refits; the row that moves it furthest is WELLSTON CENTER, the only precinct in the county above 50% Black.

The estimate degraded in exact proportion to how far it was extrapolated, which tells you it was never really an estimate about Black voters. It was a property of the line's slope. **A fifty-year-old courtroom standard, in this county, rests on one polling place** — and no report states how many precincts anchor the minority end.

15 Loosening the definition makes it worse

The intuitive fix for a method that returns nothing is to relax "homogeneous" until it returns something.

Precincts defined as homogeneous	How many qualify	Black Democratic estimate	Error against the truth
80% or more Black	0	cannot be computed	
70% or more Black	1	77.1%	-15.6
60% or more Black	1	77.1%	-15.6
50% or more Black	2	68.8%	-24.0

Each relaxation admits more white voters, who broke 7.1% Democratic, and drags the estimate down. Note that the 70% and 60% thresholds give the *identical* answer, because both capture exactly one precinct. At that point the "method" is one polling place wearing the name of a method.

There is no principled threshold; 80% is convention. A witness who reports the 70% threshold and not the 80% one has not lied about anything — the 80% returns nothing. The discretion lives in the choice of threshold, and it is invisible in the final number.

16 What the county established, and what it did not

In this electorate voting was strongly racially polarized, every method above reached that conclusion, and every one of them got the numbers wrong. Those three statements are all true at once, and the last of them is the one nobody in a courtroom ever gets to make.

Sharper: the bounds were correct and 29.9 points wide. The regression was 9.4 points off and 2.1 points past possibility, carrying an R^2 of 0.982. And the reason is nameable — its central assumption, a constant group rate, is false in this data by 10.9 points, in a direction correlated with the very variable on the horizontal axis.

None of which establishes *Gingles* preconditions two and three. Before 29 April 2026 it would have. Since *Callais* a plaintiff must show racial bloc voting **that partisanship cannot explain**, and the only outcome recorded in this file is which party's ballot a voter asked for. This chapter measures the confound; it does not break it.

Nor does any of it say how Houston County votes. A primary ballot request is not a vote, and primary electorates are smaller and more partisan than general ones. What was validated is a method, on a real electorate, for a real choice — not a county.

And the passing grade is not a clean bill of health. This is the easy case: polarization here is near-total, so bad numbers still produced the right legal answer. In a county at 55/45 the same errors would have flipped it.

Note also who is missing from every table above. Two-group ecological inference has nowhere to put Hispanic, Asian, American Indian and other voters, so they were dropped before the file was built. **They voted. The framework excluded them, not the data.**

17 Where the argument stands

Goodman versus the bounds, and then King. Goodman's ecological regression and Duncan and Davis's method of bounds appeared in the same 1953 volume of the *American Sociological Review*. They represent the two available ways to fail. Precise and possibly impossible, or guaranteed correct and possibly useless.

King's 1997 book proposed using the bounds as hard limits, with a statistical model to locate the answer inside them. It is the standard tool in current litigation. It would do better here. It would also still be a model, and it would still be uncheckable in every county but this kind.

Freedman's objection has never been answered so much as absorbed. Writing directly about voting-rights litigation, Freedman and colleagues argued that ecological regression's assumptions are not technical conveniences. They are claims about how voters behave, and the totals cannot test them. That is exactly what Figure 7 demonstrates, with the answer key in hand. The profession's response has largely been to use better models rather than to dispute the point.

The strongest defense of fifty years of practice is narrower than its defenders usually claim. These methods are reliable for *direction* and unreliable for *size*. Until 29 April 2026, *Gingles* mostly needed direction: whether the minority group votes together, and whether the majority votes as a bloc often enough to defeat its preferred candidate. That was a defensible position, and it should be stated before it is attacked. It always had a limit. Remedial questions, such as what share of a district a group needs to elect its candidate of choice, need size.

***Callais* added a second limit, and it is the worse of the two.** Direction is no longer sufficient either. What a plaintiff must now establish is direction **net of party**. That is not a more precise version of the same estimate. It is a different quantity, and neither Goodman nor the bounds was built to produce it.

This is where the tangle stops being a technicality. Race and party are so tightly linked in the American South that the two explanations look almost identical in totals. That is exactly why the requirement bites hardest in the states where Section 2 cases are actually brought. *Gingles* itself shows what satisfying it looks like. The evidence there was that Black and white voters diverged **inside** the Democratic primary, 478 U.S., at 59. That is a comparison within one party, not the between-party one run here.

And the validation is drawn from a biased sample of counties. Checking these methods requires a state that records race *and* runs partisan primaries. That is a small, mostly Southern set, and it is also where Section 2 litigation concentrates. Whether that is reassuring or circular depends on how you look at it, and both readings are defensible.

18 What one checkable county can testify to

The strength here is rare enough to be worth naming precisely: real precincts, a real election, and the answer key. Almost no dataset in this field contains the quantity being estimated, which is why the literature argues about assumptions rather than about errors. Because this one does, every method can be scored instead of merely described, and the failure can be traced to a specific false assumption rather than attributed to noise.

The limits are six. **17 precincts** is on the small side, though entirely ordinary for a county-level Section 2 case. The failure here is structural: reading past the end of the data, plus an assumption that each group behaves the same everywhere. More precincts of the same shape would fix neither.

This is a primary, not a general election. requesting a Democratic ballot in May is not voting for a Democrat in November. **Two groups only**, because everyone who is not Black or white was dropped for want of a box to put them in.

Race on the form is itself a measurement, with the same one-box problem every self-reported race question has. **King's method was not run**, only the two methods it was designed to replace, and it would do better here than either.

And **individual records are absent from the committed file** by design. The 16,024 voter records were collapsed to 17 precinct rows before anything was saved.

Four questions remain genuinely open, and the last of them is new. What a court should be handed is unsettled. The bounds are correct and decide nothing. The regression decides and is wrong. "Both, with the assumptions stated" is a procedural answer rather than a statistical one.

Whether a check in a May primary carries over to a November general, where nobody can check, requires assuming the method's *error* behaves the same in both. That is a claim about how behaviour tracks geography rather than about turnout, and nothing here tests it.

How to report the fragility has no standard at all. No convention exists for stating how many precincts anchor the minority end of a fitted line, and Figure 8 suggests it should be the first thing a reviewer asks.

And then the question *Callais* created. **What a party-controlled analysis of racially polarized voting is supposed to look like is not specified anywhere.** The Court requires one. It does not say what one is.

There are three obvious candidates. Restrict to a single party's primary. Hold the precinct's partisan baseline fixed. Or model race and party together. Each answers a slightly different question, and none has ever been scored against a truth column the way the two methods above were scored here.

Gingles itself points at the shape of an answer. The evidence there was divergence *inside* the Democratic primary rather than between the parties. But the method the law now demands is the one with the least validation behind it, which is an uncomfortable place for a standard of proof to sit.

19 Sources

Data

- Houston County, Georgia, General Primary of 21 May 2024. Party of the primary ballot requested comes from the state's voter-history file; race is self-reported on the voter registration form; precinct is from the state registration extract. All three are public records in Georgia. Working data for the 2026 Houston County redistricting matter. **Rows whose race had been filled in by BISG rather than reported by the voter were dropped** before the collapse to precinct level. The registration and voter-history files are sold by the state to anyone who orders them. https://sos.ga.gov/page/order-voter-registration-lists-and-files

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`houston_primary.csv`, `join_counts.csv`, `join_peek.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `houston_primary.csv` has, for each precinct, the number of Black and white voters and the number taking each party's ballot. Compute the Democratic share by precinct and plot it against the Black share.
- The file also has `black_dem` and `white_dem` — the truth an ecological estimate is trying to recover. Estimate it from the precinct totals alone, then check your answer.
- `join_counts.csv` shows how many rows survive each step of the join. Work out what share of the county is left at the end. Who fell out?
- Find the precincts where the ecological estimate would go furthest wrong. What do they have in common?

Literature and cases

- Goodman, L. A. (1953). "Ecological Regressions and Behavior of Individuals." *American Sociological Review* 18(6): 663–664.
- Duncan, O. D. and Davis, B. (1953). "An Alternative to Ecological Correlation." *American Sociological Review* 18(6): 665–666.
- Goodman, L. A. (1959). "Some Alternatives to Ecological Correlation." *American Journal of Sociology* 64(6): 610–625.
- King, G. (1997). *A Solution to the Ecological Inference Problem: Reconstructing Individual Behavior from Aggregate Data.* Princeton: Princeton University Press.
- Freedman, D. A., Klein, S. P., Sacks, J., Smyth, C. A. and Everett, C. G. (1991). "Ecological Regression and Voting Rights." *Evaluation Review* 15(6): 673–711.
- *Thornburg v. Gingles*, 478 U.S. 30 (1986).
- *Allen v. Milligan*, 599 U.S. 1 (2023).
- *Louisiana v. Callais*, 608 U.S. ___ (2026) (Nos. 24-109 and 24-110, decided 29 April 2026), affirming *Callais v. Landry*, 732 F. Supp. 3d 574 (WD La. 2024).
- *Allen v. Milligan*, 608 U.S. ___ (2026) (per curiam) (No. 25A1314, order of 2 June 2026).

Further reading

- Robinson, W. S. (1950). "Ecological Correlations and the Behavior of Individuals." *American Sociological Review* 15(3): 351–357.